

CAMPUS ADVISOR AI - Intelligent Student Support Assistant

1. Project Overview

Project Title

CAMPUS ADVISOR AI - An Agentic AI-Based Intelligent Student Support Assistant

Problem Statement

College students frequently need information regarding attendance requirements, academic regulations, examination rules, syllabus details, institutional policies, and general college-related queries. In many institutions, this information is distributed across multiple documents, notices, handbooks, and portals.

Finding accurate information can therefore be time-consuming and inconvenient for students. Traditional chatbots often provide predefined responses and are unable to intelligently search institutional knowledge or understand complex natural-language questions.

The problem addressed by this project is the need for an intelligent student support system that can understand student queries, retrieve relevant information from a college knowledge base, and generate accurate, contextual responses.

Campus advisor AI addresses this problem using an **Agentic AI architecture combined with Retrieval-Augmented Generation (RAG)**.

Brief Description of the Project

Campus advisor AI is an intelligent student support assistant designed to answer college-related questions through natural language interaction.

The system uses an AI agent powered by **LangGraph and LangChain** to understand user queries and determine the appropriate actions required to answer them.

A **Retrieval-Augmented Generation (RAG)** system searches a knowledge base containing college-related information such as attendance regulations, academic policies, examination rules, syllabus information, and frequently asked questions.

Relevant information is retrieved using **FAISS vector similarity search** and passed to the **Google Gemini AI model**, which generates a natural and contextually relevant response.

The project demonstrates how Agentic AI can be applied to create intelligent assistants that are capable of reasoning, retrieving information, using tools, and generating meaningful responses.

2. Objectives s Proposed Solution

Project Objectives

The main objectives of Campus advisor AI are:

- To develop an intelligent AI-based assistant for student support.
- To provide quick answers to college-related questions.
- To implement an Agentic AI architecture using LangChain and LangGraph.
- To integrate Retrieval-Augmented Generation (RAG) for knowledge-based responses.
- To retrieve relevant information from college documents using vector search.
- To reduce the time students spend searching through multiple academic documents.
- To generate natural and context-aware responses using a Large Language Model.
- To demonstrate tool usage and intelligent workflow management through an AI agent.
- To provide a scalable foundation for future college support systems.

How the Agentic AI Solution Works

Campus advisor AI follows an Agentic AI workflow where the AI agent processes a student's question and decides how to obtain the required information.

The working process is as follows:

Step 1: User Query

The student enters a question in natural language.

For example:

"What happens if my attendance falls below 75%?"

Step 2: AI Agent Processes the Query

The LangGraph-based AI agent receives and analyzes the question.

The agent determines whether it requires information from the college knowledge base.

Step 3: Tool Selection

The AI agent invokes the appropriate retrieval tool.

For college-related queries, the agent uses the knowledge-base search tool.

Step 4: RAG Retrieval

The RAG system searches the college knowledge base.

Documents are converted into vector embeddings using an embedding model. The user's query is also converted into an embedding.

Step 5: FAISS Similarity Search

FAISS performs similarity search to identify the most relevant information from the knowledge base.

Relevant document sections are retrieved.

Step 6: Context is Provided to the AI Model

The retrieved information is provided to the Google Gemini model as context.

This ensures that the response is based on the available college knowledge rather than relying only on general AI knowledge.

Step 7: Response Generation

Google Gemini generates a clear and natural response.

The response is then displayed to the student through the terminal interface.

Key Features

- Intelligent conversational AI assistant.
- Agentic AI workflow.
- LangChain integration.
- LangGraph-based agent orchestration.
- Retrieval-Augmented Generation system.

- Vector-based knowledge retrieval.
- FAISS vector store for semantic search.
- Support for attendance-related queries.
- Support for college regulations and policies.
- Support for syllabus and academic information.
- College FAQ assistance.
- Conversational and user-friendly responses.
- Environment variable configuration for secure API key management.
- Extensible architecture for adding more college documents and tools.

Implementation s Results

Technologies/Tools Used

Programming Language: Python

AI Framework: LangChain

Agent Workflow Framework: LangGraph

Language Model: Google Gemini API

RAG Components: LangChain Text Splitters and Document Processing

Vector Database: FAISS

Embedding Model: Google Generative AI Embeddings

Environment Variable Management: .env

Development Environment: Visual Studio Code

Version Control: Git and GitHub

Working Process

The implementation of Campus advisor AI began by preparing a knowledge base containing sample college-related information such as attendance regulations, academic policies, examination information, and general student guidelines.

The documents are processed and divided into smaller text segments. These segments are converted into vector embeddings and stored in a FAISS vector database.

When a student asks a question, the query is processed by the LangGraph-based AI agent. The agent can invoke the knowledge retrieval tool when relevant information is required.

The retrieval system performs semantic search on the FAISS vector store and identifies the most relevant information related to the student's question.

The retrieved information is then passed to the Gemini language model as context. The model generates a final response based on the available knowledge and the student's query.

The system then displays the answer in a clean conversational format.

The project was also configured using environment variables to securely store the Gemini API key. The .env file was excluded from the GitHub repository using .gitignore to prevent sensitive credentials from being publicly uploaded.

Screenshots/Output

The project was successfully tested through the terminal-based conversational interface.

The system displays an initial welcome message and informs users about the topics they can ask questions about. Students can then enter their queries directly into the terminal.

During testing, the system successfully answered questions related to:

- Minimum attendance requirements.
- Consequences of attendance shortages.
- Attendance eligibility conditions.
- College regulations.
- Academic information.

The system was also updated to improve the response format. Instead of displaying raw AI message structures or technical dictionary outputs, the final implementation presents the response as a clean conversational answer from Campus advisor AI.

```
campusmate-ai
-----
CampusMate AI
Intelligent Student Support Assistant
-----

Hello! I can help you with college-related questions.

Ask me about:
* Attendance
* Regulations
* Syllabus
* College FAQs
* Academic Information

Type 'help' for commands.
Type 'exit' to quit.

[Initializing CampusMate AI LangGraph Agent & RAG Vector Store...]
[System Ready. Start chatting below]

You: What's the minimum attendance percentage?

CampusMate AI: The minimum mandatory attendance requirement is **75%** across all registered theory and laboratory courses.

Here are the key details regarding the college attendance policy:

* **Standard Requirement:** You must maintain at least **75%** attendance to be eligible to sit for end-semester examinations.
* **Condonation Range (65% - 74%):** If your attendance falls between **65% and 74%** due to verified medical emergencies or official representation of the institute, you may apply for attendance condonation subject to approval from the Dean/Principal.
* **Below 65% (or below 75% without approved condonation):** Students are **ineligible** to write end-semester exams and will have to re-register for and redo the course in subsequent semesters.
* **Tracking:** Attendance status is updated weekly on the student portal.

If you know your current number of attended and total classes, feel free to share them with me, and I can calculate your exact attendance percentage for you!

You: 
```

```
data > faq.txt
1 -----
2 CAMPUS FREQUENTLY ASKED QUESTIONS (FAQ) (DEMO DATA)
3 Note: This is sample demonstration data for CampusMate AI and can be replaced with actual institution documents.
4 -----
5
6 Q: What are the central library working hours?
7 A: Central library working hours are 9:00 AM to 6:00 PM on all official working days (Monday through Saturday). The digital library and e-resources are available 24/7.
8
9 Q: Who should students contact for academic issues, marks discrepancies, or course advisement?
10 A: Students should contact their respective Academic Department / Head of Department (HOD) or designated Faculty Mentor for academic matters.
11
12 Q: Where and how can students view examination results?
13 A: Examination results are officially published and available through the online Student Information Portal (ERP) using student registration details.
14
15 Q: How can students apply for leave or on-duty (OD) permission?
16 A: Leave and On-Duty (OD) applications must be submitted via the online ERP portal at least 2 days in advance, supported by relevant medical certificates.
17
18 Q: What are the college canteen and dining facility hours?
19 A: The campus canteen is open from 7:30 AM to 7:00 PM. Breakfast is served from 7:30 AM to 9:30 AM, Lunch from 12:00 PM to 2:00 PM, and Snacks from 3:00 PM to 6:00 PM.
20
21 Q: How do I obtain bonafide certificates or transcript copies?
22 A: Bonafide requests can be applied online through the Student Portal or submitted in person at the Academic Administrative Section with supporting documents.
23 -----

[Initializing CampusMate AI LangGraph Agent & RAG Vector Store...]
[System Ready. Start chatting below]

You: What's the minimum attendance percentage?

CampusMate AI: The minimum mandatory attendance requirement is **75%** across all registered theory and laboratory courses.

Here are the key details regarding the college attendance policy:

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* **Tracking:** Attendance status is updated weekly on the student portal.
```

Results Achieved

The project successfully demonstrates the implementation of an Agentic AI system for student support.

The AI agent was able to understand student queries and retrieve relevant information from the knowledge base before generating responses. The integration of RAG improved the relevance of responses by grounding the AI output in available college information.

The project successfully integrated LangChain for AI and document processing, LangGraph for agent workflow management, Gemini as the language model, and FAISS as the vector store.

The final system provides a working prototype of an intelligent student support assistant capable of answering college-related questions in a simple conversational manner.

Conclusion s Future Scope

Project Conclusion

Campus advisor AI demonstrates how Agentic AI can be applied to improve student access to academic and institutional information.

The project combines LangChain, LangGraph, Gemini, and Retrieval-Augmented Generation to create an intelligent assistant that can understand queries, retrieve relevant information, and generate useful responses.

Unlike a basic chatbot, the system follows an agent-based approach where the AI can use tools and access a knowledge base when required. This makes the solution more suitable for information-based institutional applications.

The project successfully achieved its objective of developing a working AI-powered student support assistant that provides conversational access to college-related information.

Challenges Faced

Several challenges were encountered during the development process.

One challenge was configuring the correct language model because an initially selected Gemini model was no longer available. This was resolved by updating the project configuration to use a supported Gemini model.

Another challenge was dependency installation, as AI frameworks such as LangChain, LangGraph, FAISS, and embedding-related packages require multiple dependencies.

The initial AI response format also displayed raw message structures and technical dictionary content. The response handling logic was improved so that only the actual AI-generated text is displayed to the user.

The project also experienced additional processing time while loading embedding models and retrieving information from the vector store. These processes are necessary for semantic search but can be optimized further in future versions.

Finally, API key security was addressed by using a .env file and excluding it from the GitHub repository through .gitignore.

Future Enhancements

The following enhancements can be implemented in future versions:

- Development of a web-based graphical user interface.
- Student login and authentication system.
- Integration with real college databases.
- Upload and processing of official PDF documents.
- Support for multiple departments and courses.
- Personalized responses based on student profiles.
- Integration with attendance management systems.
- Examination timetable and result information.
- Notification and reminder features.
- Voice-based interaction.
- Multilingual support.
- Integration with college websites and portals.
- Deployment as a cloud-based application.
- Mobile application development.
- Addition of more AI tools for advanced student support.

References

1. [LangChain Documentation](#)
2. [LangGraph Documentation](#)
3. [Google Gemini API Documentation](#)
4. [FAISS Documentation](#)
5. [Python Documentation](#)
6. GitHub Repository : <https://github.com/sakthivel2k/campus-advisor-AI.git>